

PET Count Rate Performance

Program Documentation

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1 Overview

The **Count Rate Performance** program is a simplified PET uniformity workflow designed to compute average standardized uptake value (SUV) within a fixed 180 mm diameter region of interest (ROI) across three user-selected slices of a PET phantom.

This script provides a reproducible and Siemens-aware implementation of SUV calculation, intended to evaluate system performance under varying count rate conditions (e.g., low-activity versus high-activity acquisitions). It generates a concise summary of mean SUV values and a visual overlay for quick inspection.

2 Purpose

This program was developed to:

- Quantitatively assess PET system count-rate linearity and uniformity stability.
- Verify SUV constancy across reconstructed frames or injected activity levels.
- Provide a standardized image visualization (0–3 SUV window) for inclusion in PET QA reports.
- Produce consistent, quantitative data for comparison against baseline or acceptance measurements.

3 Program Workflow

3.1 Step 1: Input and Output Selection

The program prompts the user to:

1. Select a folder containing PET DICOM images.
2. Choose an output directory and specify a subfolder name for results.

All resulting CSV and image outputs are written to this subfolder.

3.2 Step 2: DICOM Loading and Preprocessing

The DICOM series is read using `pydicom`, and only valid 2D PET images are retained.

Each image undergoes preprocessing:

- DICOM `RescaleSlope` and `RescaleIntercept` are applied to reconstruct activity concentration (Bq/mL).
- Siemens PET datasets with `Units = BQML` are recognized as already in Bq/mL, so no slope is re-applied.

- The images are sorted in slice order based on the `ImagePositionPatient` tag.

The program extracts pixel spacing and ensures consistent matrix dimensions across all slices.

3.3 Step 3: Slice Selection

A grid of preview images is displayed. The user must click exactly three slices corresponding to the uniformity evaluation frames.

Each preview shows a scaled grayscale display of the reconstructed PET data with an automatically applied field of view of 250 mm.

3.4 Step 4: Center Definition

Once the slices are selected, the program prompts the user to select the phantom center on the first chosen slice using a cursor overlay. This single center point (in millimetres) is then reused for all three slices to ensure identical ROI placement and consistent comparison across frames.

3.5 Step 5: SUV Calculation

Each slice is processed through the internal SUV calculator (`suv_from_ds`), which performs the following operations:

- Converts pixel values to activity concentration (Bq/mL) using DICOM scaling factors.
- Retrieves the injected dose (`RadionuclideTotalDose`) and half-life (`RadionuclideHalfLife`) from the `RadiopharmaceuticalInformationSequence`.
- Decays the injected dose to the reference time (`SeriesDate/Time`) unless the DICOM tag `DecayCorrection` equals `ADMIN`, in which case the activity at injection is used.
- Applies patient body weight (or optionally LBM, IBW, or BSA) as the normalization factor.

The resulting SUV image is windowed from 0 to 3 SUV and stored for display.

3.6 Step 6: ROI Placement and Measurement

A circular ROI of diameter 180 mm is automatically drawn around the selected center. For each of the three slices:

- The mean SUV within this ROI is computed.
- If the ROI extends outside image boundaries, a warning is issued.

These mean values are collected for export and labeling.

3.7 Step 7: Output Generation

The following files are created:

- **Count_Rate_Performance.csv**: Contains the slice indices and corresponding mean SUV values.
- **Count_Rate_Performance.png**: A 1×3 image panel of the selected slices, each annotated with the text “Mean = X.XX SUV”.

All display panels use a standardized grayscale window of 0–3 SUV (white = 0 SUV, black = 3 SUV) to ensure consistent visual interpretation across datasets.

4 SUV Calculation Summary

The program's SUV computation follows the equation:

$$\text{SUV} = \frac{C_{\text{voxel}} \times M}{A_{\text{ref}}}$$

where:

- C_{voxel} = activity concentration in Bq/mL,
- M = patient mass normalization (in grams, default = body weight),
- A_{ref} = injected activity decay-corrected to the reference time (typically the start of acquisition).

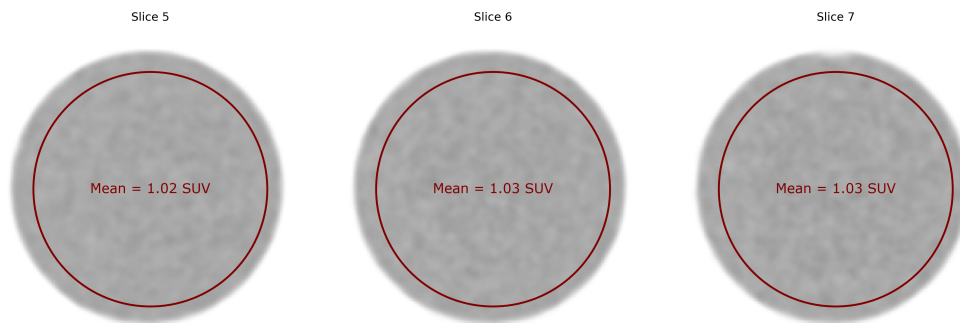
For Siemens datasets:

- If `Units = BQML`, pixels are already in Bq/mL.
- If `DecayCorrection = ADMIN`, no decay correction is applied ($A_{\text{ref}} = A_0$).
- For other cases, $A_{\text{ref}} = A_0 e^{-\lambda t}$, where $\lambda = \ln 2 / T_{1/2}$.

5 Visualization

The composite PNG shows all three uniformity slices side-by-side:

- Each panel includes the circular ROI boundary and the corresponding mean SUV value.
- The layout ensures quick comparison of spatial uniformity across frames.



6 Interpretation

The count-rate uniformity results provide insight into the PET system's ability to maintain consistent image quantification across varying activity levels. Deviations in mean SUV across the three frames may indicate:

- Detector dead-time effects at high count rates.
- Incomplete decay correction or calibration drift.
- Systematic reconstruction scaling differences.

7 How to Use

1. Run `5..Count_Rate_Performance.py`.
2. Select the PET DICOM folder for input.

3. Choose an output directory and assign a subfolder name.
4. Click exactly three slices in the preview window.
5. Click the phantom center on the first selected slice.
6. Wait for results — both the CSV and PNG outputs are automatically generated.

8 Conclusion

The **Count Rate Performance** program provides a fast, standardized method for evaluating PET scanner linearity and uniformity across multiple frames. By employing Siemens-aware SUV correction and consistent ROI placement, it ensures reproducible measurements aligned with NEMA standards and clinical QC expectations.